MIL-D-83935B
28 September 1988
SUPERSEDES
MIL-D-83935A(USAF)
26 September 1973

MILITARY SPECIFICATION

DISHWASHING SYSTEM (COMMERCIAL), MULTIPLE-TANK, CONTINUOUS OVAL CONVEYOR-TYPE

1. SCOPE

1.1 Scope. This specification covers a commercial, multiple-tank, conveyor, spray-type dishwashing system with an oval-shaped, horizontal table-conveyor system.

1.2 Classification. The dishwashing system shall be one of the following models, types, styles, and rotations, as specified (see figure 1 and 6.2).

Model 1 - Access doors and controls mounted on inside of oval conveyor
Model 2 - Access doors and controls mounted on outside of oval conveyor

Type I - Steam Heated
Type II - Electrically Heated

Style A - Includes tray rail and table-mounted garbage disposal machine.
Style B - Includes food waste trough and garbage disposal machine mounted in center trough.
Style C - Includes tray rail. Food discharged to garbage can at disposal location in table.
Style D - Includes food waste trough. Food discharged to garbage can at disposal location in trough.

Rotation CW Conveyor moves clockwise when viewed from above.

Rotation CCW Conveyor moves counter-clockwise when viewed from above.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7320

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

1.3 Part number. Dishwashing systems procured under this specification shall be identified in accordance with the following part numbering system.

M83835 - X - X - X
- M83835 — Military Specification Number
- X — Type Designator (see 1.2)
- X — Style Designator (see 1.2)
- X — Model Number Designator (see 1.2)

Definitive Specification Part Number

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

OO-D-1388	- Dishwashing Machines, Commercial (Rackless Conveyor)
OO-D-1390	- Dishwashing Machines, Single Tank (Rack, Manual) and Conveyor, and Double Tank Commercial (Rack, Conveyor), Commercial
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting
TT-V-119	- Varnish, Spar, Phenolic-Resin
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-P-40	- Packaging and Packing of Hand Tools

MILITARY

MIL-V-3	- Valves, Fittings and Flanges (Except for Systems Indicated Herein), Packaging of
MIL-C-104	- Crates, Wood; Lumber and Plywood Sheated, Nailed and Bolted
MIL-P-116	- Preservation, Methods of
MIL-G-15840	- Garbage Disposal Machines, Commercial and Household
MIL-E-16298	- Electrical Machines Having Rotating Parts and Associated Repair Parts: Packaging of
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-882 - Systems Safety Program: Requirements for
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY OF SANITARY ENGINEERS (ASSE)

ASSE STD No 1004 - Dishwashers

(Application for copies should be addressed to the American Society of Sanitary Engineers, P.O. Box 9712, Bay Village, OH 44140.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 Standard Practice for Commercial Packaging

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

NATIONAL SANITATION FOUNDATION

NSF STD No 3 - Commercial Spray-Type Dishwashing Machines - A Seal of Approval Listing of Food Service Equipment

NSF STD No 5 - Commercial Gas Fired and Electrically Heated Hot Water Generating Equipment (for Food Service Establishments Using Dishwashing Machines)

(Copies should be obtained from the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

UNDERWRITERS' LABORATORIES (UL)

UL 921 - Commercial Electrical Dishwashers

(Copies should be obtained from the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The dishwashing system shall consist of a multiple-tank, conveyor, rackless or rack-type dishwashing machine, a direct-connected, horizontal, conveyor-table, a tray rail or food waste trough, specified associated equipment and controls, wiring, and piping necessary to have a complete, ready to use system. The tray rail or food waste trough shall be detachable from the main system. The dishwashing machine and conveyor-table, when assembled, shall form an oval shaped dish handling system generally similar to the applicable layout of figure 1. The overall length and width of the system, excluding tray rail or food waste trough, shall not exceed 17 feet and 7 feet, respectively. The system shall be designed to receive dirty dishes directly from the dining facility and to mechanically convey the items from the conveyor loading area, through the dishwashing machine, by a clean dish removal area, and back to the conveyor loading area. The provisions for garbage disposal and the selection of tray rail or food waste trough shall be in accordance with the style specified. The location of access doors and controls, the rotation direction, and whether or not racks and/or a silver soak sink are to be provided, and/or electromagnetic interference suppression is required shall be in accordance with the model specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.3 Codes and standards. The dishwashing system shall comply with the applicable requirements of Underwriters' Laboratories Standard No. 921, and the applicable requirements of National Sanitation Foundation Standard No. 3 (see 4.5, 4.5,1, and 4.5.2).

3.3.1 Data name plate. The dishwashing system shall be furnished with a data name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for; (a) Methods of applying, (b) Identification tags, (c) Information not required, and (d) Optional marking information shall not apply. The data name plate shall be made of minimum 20 gauge corrosion resisting metal, and attached to the dishwashing system by rivets, screws, or by welding, in such a manner as to meet the applicable National Sanitation Foundation requirements for this equipment. The plate shall bear the following information which shall be stamped, engraved, or applied by photosensitive means:

National Stock Number
Procurement Instrument Identification Number
Specification Data
Manufacturer's Name, Address, and Telephone Number
Supplier's Name, Address, and Telephone Number (List only if different from Manufacturer)
Manufacturer's Model Number
Government Approved Manual Number (see 6.2)

Each plate shall be placed so as to be readily visible to the operator during normal operations and use, and so as to not adversely affect the life or utility of the dishwashing system.

3.4 Standard product. The dishwashing systems shall be the manufacturer's current standard commercial model which meets or exceeds the requirements of this specification and is identifiable by the manufacturer's established model number or equivalent designation. The systems shall be new and unused except to the extent required to perform the tests specified herein.

3.5 Interchangeability. All dishwashing systems of the same type and style furnished under a specific contract shall be identical to the extent required to permit interchangeability of replaceable parts, components, and assemblies.

3.6 Material. Material for the dishwashing system shall be as specified herein. Parts for which materials are not definitely specified shall be fabricated of materials acceptable under NSF STD No 3, or, if not covered therein, the materials shall be of corrosion resistant material. No mild or galvanized steel shall be used for any structural support component. Materials shall be free from any defects that would contribute to premature failure of the machine or would reduce or impair the usability of the unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

3.7 Performance. The dishwashing system and associated table-conveyor system shall have a rated capacity as listed in the latest issue of the NSF Seal of Approval Listing of Food Service Equipment of not less than 270 dish racks (20-inch by 20-inch) or an equivalent number of dishes per hour with a conveyor speed not exceeding the rates established in the NSF Seal of Approval Listing as applicable to the model being furnished. When operating at rated capacity, the system shall comply with the applicable functional requirements of NSF STD No 3, including cycle time and water temperatures, pressures, volumes, and spray patterns for the pumped wash, pumped rinse, and final rinse.

3.8 Dishwashing system. The dishwashing system shall be similar to the machine identified in NSF STD No 3 as a circular, conveyor-type. The hoods, sliding doors, scrap recepatacles and tanks shall be fabricated of type 301, 302, or 304 stainless steel having a thickness of not less than 16 gage. Sliding doors and scrap receptables, if die formed, shall be fabricated of type 301, 302, or 304 stainless steel having a thickness of not less than 18 gage. The system shall be mounted on stainless steel, adjustable feet conforming to NSF STD No 3. Vertical sliding doors not less than 16 inches wide shall be furnished on each wash and rinse compartment and shall be equipped with suitable latching mechanism to hold the doors in the fully opened position. The machine shall consist of the following compartments:

a. Prewash
b. Pumped wash
c. Pumped rinse
d. Final rinse

3.8.1 Prewash compartment. A prewash compartment meeting the requirements of NSF STD No 3 shall be furnished.

3.8.2 Pumped wash compartment. The pumped (power) wash compartment shall be equipped with stainless steel scrap screens easily removed from the compartment. The compartment shall be equipped with an overflow pipe, a makeup water supply connection with manually-operated control valve, and a hand-or-foot operated drain valve having a nominal pipe size of not less than 1 1/2 inches. The water in the tank shall be automatically maintained by a thermostatic control at a predetermined temperature not less than 150°F, by steam or by electrical heating elements in accordance with the type specified (see 1.2). A dial type thermometer shall be mounted on the front of the machine to indicate the temperature of the water in the spray manifold. The wash section shall include an electrical motor-driven pump rated to meet or exceed the applicable performance requirements of NSF STD No 3 with respect to volume, pressure, cycle time, and spray pattern for recirculation of water from the tank to the spray manifolds. The spray boxes or spray arms shall be readily removable from the spray manifolds without use of tools, and spray arms shall be provided with readily removable end caps to facilitate cleaning.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

3.8.3 Pumped rinse compartment. The pumped (power) rinse compartment shall conform to the requirements specified in 3.8.2 for the wash compartment, except that the water in the tank shall be automatically maintained by a thermostatic control at a predetermined temperature not less than 160°F.

3.8.4 Final rinse. A final rinse compartment shall be furnished which complies with the requirements of NSF STD No. 3.

3.9 Conveyor table. The conveyor table shall be constructed of type 302 or 304 stainless steel having a thickness sufficient to carry design loads but shall not be less than 14 gauge. The table shall be of welded construction except for screwed or bolted joints necessary to meet the maximum dimension requirements of this paragraph, shall be water tight, shall drain toward the prewash section, and shall otherwise be constructed to meet the cleanability and sanitation requirements of NSF. The table shall be supported on stainless steel legs having an outside dimension of not less than 1 5/8 inches. The legs shall be preassembled with suitable cross bracing and shall be furnished with NSF approved adjustable feet. The table shall include a mechanical conveyor system designed to continuously transfer dishes or racks from the dish loading area through the machine to the clean dish area at the rate of 6 to 8 feet per minute. The conveyor system shall include suitable means for sweeping food waste and liquids into either the garbage disposal machine or the prewash scrap baskets. The conveyor table at the dish loading area shall be equipped either with a tray rail or food waste trough in accordance with the style specified (see 1.2). The conveyor table shall be sectionalized for ease of shipping and no section shall exceed the following maximum dimensions:

Length - 10 feet, 6 inches
Width - 3 feet, 6 inches
Height - that which will permit transport though a 6' 8" door by manpower and light hand operated equipment

A back splash of not less than 18 gage stainless steel shall extend approximately 12 inches above the inner edge of the conveyor table, except at drive mechanisms, etc.

3.10 Controls and power.

3.10.1 Central controls. The system shall be furnished with a control cabinet meeting the requirements of Underwriters' Laboratories, Inc. Standards. The cabinet shall be mounted in a readily accessible location, and shall contain all controls necessary for operation of the machine, including but not limited to the following:

a. Individual thermal overload switches for each pump, tank heat circuit, and conveyor drive shall be readily accessible in the control cabinet.

b. An automatic device to conserve wetting agent, detergent, final rinse, etc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

c. An automatic device with means of manual reset to shut down the conveyor in case of a jam or other mechanical malfunction.

d. A timing device adjustable from 0 to 5 minutes to operate the disposal and its water supply after the system has been stopped.

e. Acess doors and controls shall be placed on the inside of the oval conveyor for model 1 or on the outside of the oval conveyor for model 2.

3.10.2 Remote controls. A minimum of two remote start/stop manual switches shall be at locations convenient for use by the operator.

3.10.3 Electrical power. Unless otherwise specified (see 6.2), the system shall be designed for operation on a 208 volt, 3 phase, 60 hertz power supply. Control circuits shall operate on single-phase power not exceeding 120 volts AC.

3.10.4 Prewiring of controls and componets. All electrical components and controls shall be prewired to the extent that only the reconnection of sectionalized units and connection to each power source is required upon installation.

3.10.4.1 Circuits for type II machines. Type II machines shall be prewired to include the following circuits:

a. Motors and controls
b. Tank heating
c. Booster heating

3.11 Piping. All internal piping on the machines shall be complete up to the point where connections to water, drain, and for type I machine, steam lines, will be made at the site. The piping system shall include all necessary piping, fittings, vacuum breakers, line strainers, hand-operated valves, and automatically controlled solenoid valves. Valves for manual adjustment of flow rates shall be of the globe, angle or ball valve design type. All piping shall be in accordance with ASSE STD No 1004.

3.12 Details of components.

3.12.1 Heating. Steam heated, type I machines shall be equipped with steam injectors in the power wash and power rinse tanks for tank heating and a steam booster heater for the final rinse water. Electrically heated, type II machines shall be equipped with electrical immersion heaters in the power wash and power rinse tanks for tank heating and an electric booster heater for the final rinse water. Tank heaters shall be sized and rated to meet the temperature requirements of NSF STD No 3 and shall be thermostatically controlled to automatically maintain the specified tank water temperatures. Tank heaters for type II classification shall be complete with low water cutoff protection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

3.12.2 Booster heaters. Booster heaters shall automatically maintain the required final rinse water temperature without producing steam either within heater or the piping connected to the heater. Booster shall be mounted under the conveyor table to form a component of the machine and shall be furnished complete with electric thermostat, pressure relief valve, pressure regulator and all other related controls and components necessary to assure safe operation. Steam booster for type I machines shall be factory preplumbed and prewired suitable for operation with steam pressures ranging from 15 psig to 40 psig at the machine. Electric booster for type II machines shall be factory preplumbed and prewired with high temperature limit switch for protection. Booster heaters shall have sufficient capacity, with a 60°F temperature rise, to meet the volume and temperature requirements of NSF STDs No 3 and No 5.

3.12.3 Pumps. Pumps for the prewash, power wash, and power rinse sections shall be furnished with stainless steel shafts, stainless steel, nickel alloy or cast iron impellers, and a removable inspection and clean-out plate, or a readily accessible strainer screen over the pump sump.

3.12.4 Garbage disposal machine. When style A or style B dishwashing machines are specified, a gargage disposal machine shall be furnished with each unit. The garbage disposal machine shall conform to type I (commercial) of MIL-G-15840. The machine shall be rated at not less than 3 horsepower and shall have a capacity of at least 1600 pounds per hour. The disposal machine shall be suitable for installation in the locations indicated in figure 1 in accordance with the style specified.

3.13 Accessories.

3.13.1 Detergent dispenser. Means for connecting an automatic detergent dispenser as specified in OO-D-1388 or OO-D-1390 shall be provided on the wash compartment of the machine.

3.13.2 Rinse injector. Means for connecting a drying agent injector as specified in OO-D-1390 or OO-D-1388 shall be provided on the final rinse compartment of the machine.

3.13.3 Racks. When specified (see 6.2), fifteen combination glass-dish racks shall be furnished with each machine. The racks shall be nominal 20-inch by 20-inch racks and shall be the manufacturer's standard racks suitable for use with the conveyor system being furnished.

3.13.4 Silver soak sink. When specified (see 6.2), a portable, caster-mounted soak sink for silverware shall be furnished. The sink shall be designed to accomodate standard silverware racks and shall be approximately 22 inches by 22 inches by 6 inches minimum deep. The sink shall be fitted with a drain hose. The sink and dolly-stand shall be of stainless steel construction. The height to the top of the sink shall be 34 inches ± 1 inch from the floor. Casters shall be of the swivel type with wheels not less than 4 inches in diameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

When a soak sink is specified, two silverware racks shall be furnished with each sink. The stand shall include a bottom shelf for storing extra silverware racks.

3.14 Treatment and painting. All exterior surfaces of the dishwasher except stainless steel surfaces, shall be treated and painted in accordance with the manufacturer's standard practice.

3.15 Identification marking. The dishwashing machines shall be marked for identification in accordance with MIL-STD-130.

3.16 Instruction plate. An instruction plate of corrosion-resistant metal or chrome-plated brass shall be attached to each machine at a height readily visible to the operator. The plate shall provide numbered, step-by-step instructions for operating and cleaning the dishwasher. Numbers corresponding to those on the instruction plate shall be attached on the machine at a location adjacent to the operating position. The instruction plate shall list the required water temperature for each section (prewash, wash, power rinse, final rinse).

3.17 Electromagnetic interference (EMI) suppression. When specified (see 6.2 and 6.4) the dishwashing system shall conform to the EMI control requirements of MIL-STD-461 for the equipment class specified in the procurement document.

3.18 Fungus treatment. When specified (see 6.2), the dishwashing machines shall be treated or constructed to resist fungus. Materials that are nutrients for fungi, if used, shall be treated with a fungicidal agent.

3.19 Safety. All equipment furnished under this specification shall be analyzed for category III and category IV hazards as defined in MIL-STD-882. Any hazard so identified shall be brought to the attention of the procuring agency with recommended action for satisfying safety requirements. Action order of precedence shall be as specified in MIL-STD-882.

3.20 Workmanship. The dishwashing system shall be neatly finished and all burrs and rough edges shall be removed. Welds shall be sound, smooth, and free from pits, fractures, burn through or fissures. Bolt and rivet holes shall be accurately punched or drilled and shall have burrs removed. Lock washers shall be provided in accordance with standard commercial practice, and all bolts, nuts and screws shall be drawn tight. Rivets shall be tightly driven and fully peened. Electrical wiring shall be provided with adequate slack to relieve strain, and the wiring shall not be abraded, poorly joined or loose at the terminals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in table I and shall be tested as specified in 4.5. The presence of any defect or failure of any test shall be cause for rejection of the first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

TABLE I. First article defects

Defect	Classification Major
Missing components	X
Tray rail and food waste trough not detachable	X
Dishwashing machine and conveyor table not oval shaped	X
Length not as specified	X
Unable to receive dirty dishes directly from dining facility	X

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 End item examination. Machines furnished under this specification shall be examined during and after fabrication in accordance with the manufacturer's established quality control program. The quality control methods and procedures shall assure compliance with the requirements of this specification and referenced documents throughout all areas of contract performance including the recipt and identification of material, fabrication, and final assembly of the machine. The quality control program shall be subject to surveillance by the procuring agency or designated representative thereof.

4.4.3 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

Examine	Defect
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.4 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Certification compliance examination. Certifications certified test reports, or listing marks for codes and standards, as applicable, submitted in accordance with 3.3, shall be examined and validated as proof of compliance.

4.5.1 Underwriters Laboratories, Inc. (UL). Acceptable evidence of meeting the applicable requirements of the Underwriters Laboratories standards specified in 3.3 shall be the UL label, a UL listing mark, or a certified test report from a recognized independent laboratory, acceptable to the Government, indicating that the dishwashing system and conforms to the applicable requirements of the specified UL standards (see 3.3).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

4.5.2 National Sanitation Foundation (NSF). Acceptable evidence of meeting the applicable requirements of the National Sanitation Foundation standards specified in 3.3 shall be one of the following:

a. NSF STD No 3 - The official NSF symbol on the equipment or a listing in the latest issue of the NSF Seal of Approval Listing of Food Service Equipment.

b. UL - A listing of the latest issue of the UL Electrical Appliance and Utilization Equipment List or the official UL marking on the equipment authorized under the UL's Re-examination Service.

c. ASSE STD 1004 - Certification that the dishwashing system is in compliance with American Society of Sanitary Engineers' Standard 1004.

4.6 Methods of inspection.

4.6.1 Operational test. Each machine shall be operated for at least one hour without dishes to determine compliance with this specification. The tests shall verify that the assembly of component parts is complete and of proper inter-relationship; that controls function properly; that tanks and piping systems do not leak; and that the overall performance of the unit is satisfactory.

4.6.2 Capacity test. Racks shall be placed on the conveyor for a period of not less than 10 minutes to determine compliance NSF STD No. 3 (see 3.7).

4.6.3 Electromagnetic interference tests. When EMI suppression is specified, the first article sample shall be tested in accordance with MIL-STD-462, for the specified class of equipment (see 6.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly of components and parts. The dishwashing system shall be disassembled into components and parts meeting the requirements of this specification. Removed nuts, bolts and washers shall be secured to mating parts.

5.1.1.2 Unpainted ferrous metal surfaces. All unpainted ferrous surfaces shall be coated with P-14 preservative oil in accordance with MIL-P-116. The breaking, pulverizing, and expelling section of the garbage disposal machine shall be coated with type P-9 preservative oil in accordance with MIL-P-116.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

5.1.1.3 Motors and electrical parts. The motors and electrical parts shall be preserved and unit packed in accordance with MIL-E-17555 and MIL-E-16298, as applicable.

5.1.1.4 Drive belts and pulleys. Drive belts shall be removed or the belt tension shall be relieved. If belts are removed they shall be preserved individually or in sets by method 1C-1 in accordance with MIL-P-116. Pulley grooves shall be coated with lacquer resisting primer conforming to TT-P-664.

5.1.1.5 Threaded connections. Inlet and outlet threaded connections shall be sealed with threaded metal or plastic caps or plugs.

5.1.1.6 Gages and thermometers. When gages or thermometers are removed, openings shall be sealed with commercial plugs or caps. Gages and thermometers shall be preserved in accordance with method III of MIL-P-116.

5.1.1.7 Valves and fittings. Valves and fittings shall be preserved and unit packed in accordance with the level A requirements of MIL-V-3.

5.1.1.8 Nameplates and instruction plates. Nameplates and instruction plates shall be coated with spar varnish conforming to TT-V-119.

5.1.1.9 Racks. Racks shall be unit packed in one or more snug-fitting fiberboard boxes conforming to PPP-B-636.

5.1.1.10 Disassembled components and parts, repair parts, and tools. Disassembled components and parts, repair parts and tools shall be preserved and unit packed in accordance with MIL-E-17555, MIL-V-3, MIL-E-16298, and PPP-P-40 and with the applicable methods described in PPP-B-636 for parts not specifically covered in the aforementioned documents.

5.1.1.11 Intermediate packaging. Disassembled components, parts and tools shall be packaged in one or more fiberboard boxes conforming to PPP-B-636, class 1. Contents shall be cushioned and braced to prevent movement within the boxes.

5.1.2 Commercial. The dishwashing systems shall be disassembled into components and parts meeting the requiremnts of this specification. The component shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or Commercial as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

5.2.1 Level A packing. The dishwashing system components as disassembled (see 5.1.1.1) including the garbage disposal machine and silver soak sink, when furnished, preserved as specified in 5.1, shall be packed in type I, class 2, style A crates conforming to MIL-C-104. No crate shall exceed the following overall dimensions:

Length - 11 feet, 9 inches
Width - 4 feet, 0 inches
Height - 7 feet, 6 inches

Contents shall be cushioned, blocked, braced, and anchored in accordance with MIL-STD-1186 and MIL-C-104.

5.2.2 Commercial packing. The disassembled dishwashing system (see 5.1.2) including the garbage disposal machine and silver soak sink, when furnished, shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. No container shall exceed the overall dimensions of 5.2.1. Containers and packing shall comply with ASTM D 3951.

5.2.3 Marking. In addition to any special marking required by the contract or order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

6. NOTES

6.1 Intended use. The dishwashing system is intended for use in installations requiring large scale service of food. The system, capable of operation by one person, shall uniformly wash, rinse, and sanitize eating and drinking utensils by spraying them from above and below with an aqueous detergent solution and rinse water.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Model, type, style, and rotation required (see 1.2 and Figure 1).
c. When a first article is required (see 3.2).
d. When electrical power supply will be other than specified (see 3.10.3).
e. When racks are required (see 3.13.3).
f. When a silver soak sink is required (see 3.13.4).
g. When electromagnetic interference suppression is required (see 3.17).
h. When fungus treatment is required (see 3.18).
i. Level of preservation and packing required (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

6.3 Electromagnetic interference. When EMI suppression is required in accordance with the specified model, it is recommended that the procurement document specify the following equipment class according to the expected end use location of the system being procured:

End Use Location	Equipment Classification
Permanent Garrison Locations	Class C3, Group 1 for Critical Areas.
Submarine	Class A5, Equipment Category: 'All others not listed herein'.
Shipboard (surface)	Class A4, Equipment Category: 'All others not listed herein'.
Unknown Locations	Same as Shipboard (surface).
All other Locations	Class A3, Equipment Category: 'All others not listed herein'.

6.4 Subject term (key word) listing.

Commercial dishwasher
Dishwashing system

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Army - GL
Air Force - 99

Review activity:

Air Force - 84

Preparing activity:

Army - GL

Project Number: 7320-0823

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

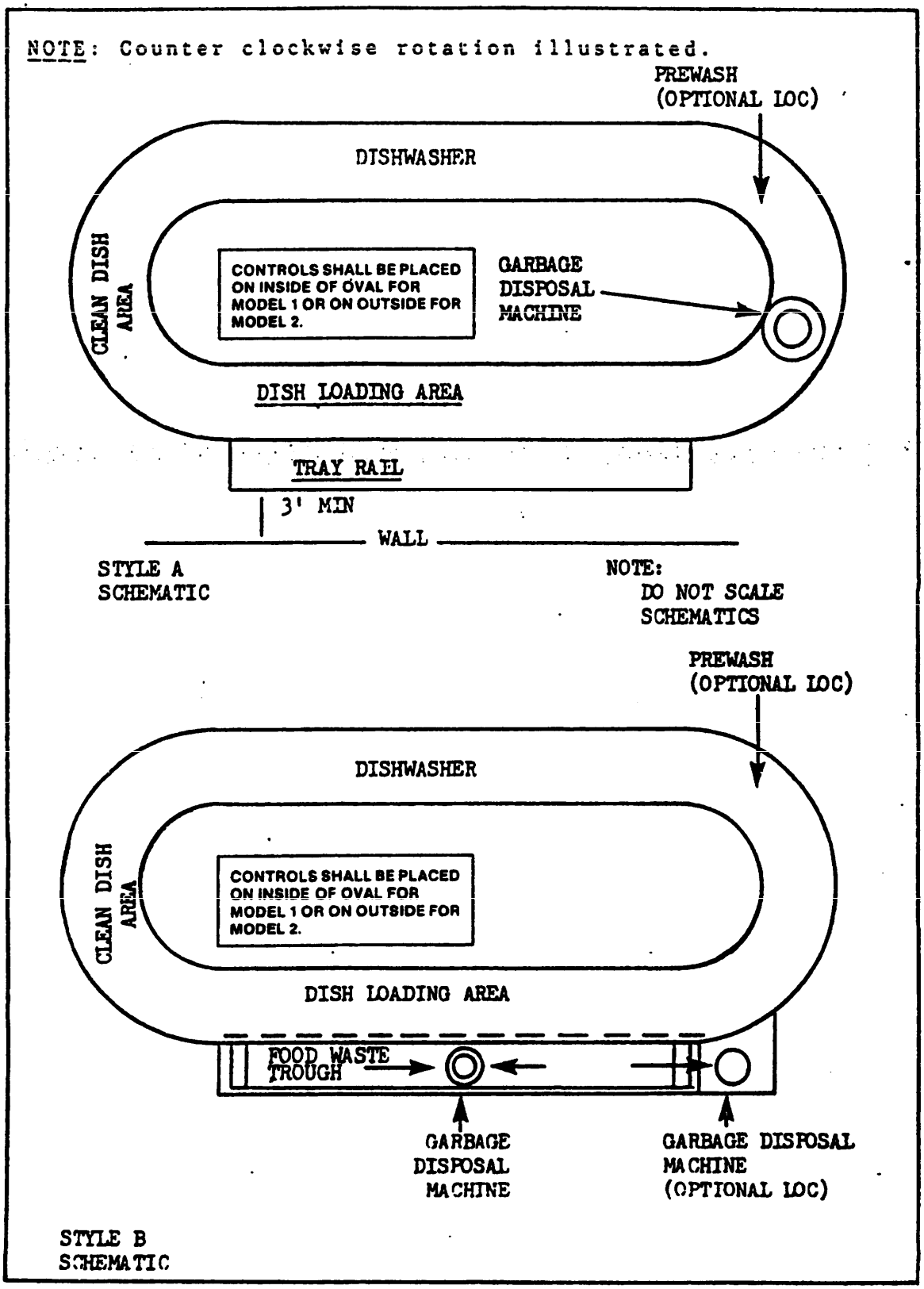


FIGURE 1. Schematic Layouts for Style A and Style B Dishwashing Systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-83935B	Dishwashing System (Commercial), Multiple-Tank, Continuous Oval Conveyor-Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. ~~Paragraph Number and Wording~~:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for ... in submitting comments and suggestions for improvements. All users of military standardization documents are invited to ...ide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: **This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.**

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

U.S. ARMY NATICK RESEARCH, DEVELOPMENT
and ENGINEERING CENTER
ATTN: STRNC-ES
Natick, MA 01760-5014

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
U.S. Army Natick Research, Development
and Engineering Center
ATTN: STRNC-ES
Natick, MA 01760-5014

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.